ULTRAMARIN RESEARCH SEMINAR · AUGUST 2026

Count Each Idea Once

Building robust composites from correlated alpha signals

Hashim Almodamagha · Master of Financial Engineering, UC Berkeley Haas

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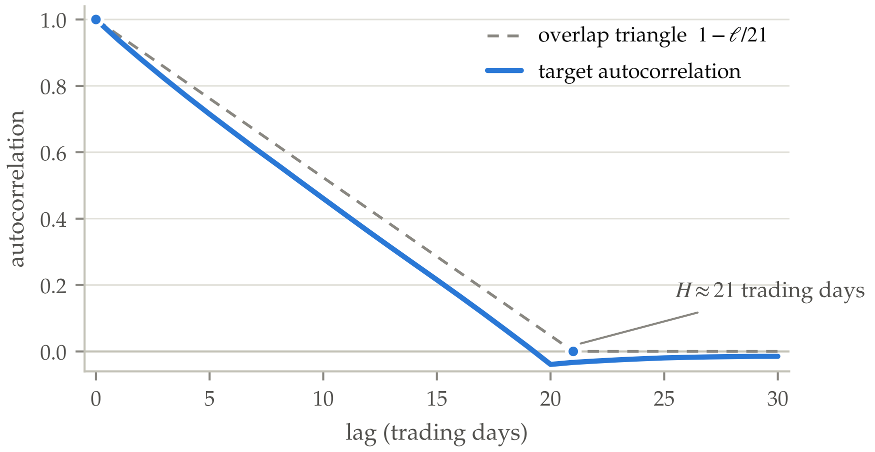
Four anonymized feature classes, one overlapping target

	VARIANTS	IN-SAMPLE DAYS	ROWS	HOLD-OUT DAYS	PC1 VAR. SHARE
class 1	16	3,130	1.51M	2,195	0.48
class 2	10	1,565	0.61M	2,195	0.51
class 3	6	1,565	0.61M	2,195	0.53
class 4	20	3,130	1.51M	2,195	0.30

Variants arrive as within-day cross-sectional ranks (~390–480 names/day); the target is a residualized cumulative forward return over ~21 trading days, sampled daily. Blind hold-out = the immediately following period, delivered only after the in-sample study was frozen.



FLAG THIS NOW

Class 4's diffuse spectrum (PC1 share 0.30 vs ~0.5 elsewhere) makes it the one genuinely multi-factor cluster. It becomes the protagonist of this talk.

Consecutive days' targets share 20 of 21 forward days, so autocorrelation decays linearly to zero at lag 21. **This one property sets the 21-day purge window and the 25-lag Newey–West errors for the entire study.**

THE ENDING, UP FRONT

Three claims. The rest of the talk is how they were earned

1

Each cluster is worth what it's worth

Every class has a hard accuracy ceiling that equal weighting essentially attains. Trees, PLS, GLS, tuned penalties cannot move it. After multiplicity correction, **nothing beats the benchmark on raw IC**, including our own headline win.

2

Deduplication is the one idea that survives every test

Merge near-duplicates into blocks and weight the blocks equally. It never loses anywhere, holds up best when the feature set is perturbed, and delivers the study's only significant win, exactly where duplication bias binds.

3

The real edge is signal quality, not IC level

The selected composite sits at or within noise of all four ceilings, which no rival method manages. Where it differs from the benchmark it **decays slower and trades at 35–40% lower turnover**.

Ground rule: every claim ships with the test that could have falsified it: purged CV, paired Newey–West, a blind hold-out, and a family-wise multiplicity correction at the end.

THE PROBLEM

A factor is never one feature. It's a cluster of correlated variants

Different lookbacks, normalizations, residualizations: each variant is a noisy measurement of the same latent signal. They must become one composite.

The benchmark everyone reaches for: rank each variant cross-sectionally, flip by the sign of its training relationship with the target, average.

$$F_t = \frac{1}{N}\sum_{j=1}^{N} s_j \, r_{j,t}$$

It is much harder to beat than it looks. But it has one structural flaw: **it weights definitions, not ideas.**

The failure mode

If 8 of 10 variants are re-implementations of "cumulative return," the average *is* that one idea, eight times over, and the other two ideas are diluted to noise.

THE OBJECTIVE

A systematic, statistically sound way to weight variants by *uniqueness and information content*, rather than by how many near-duplicates happened to land in the cluster.

One fixed harness, defined before any comparison ran

metric	definition	reads as
mean IC	$\overline{\mathrm{IC}_t}$	average daily predictive strength
IC-IR	$\overline{\mathrm{IC}}/\mathrm{std}(\mathrm{IC}_t)$	risk-adjusted consistency
Newey–West t	mean IC over its HAC (25-lag) s.e.	significance under serial correlation
signal decay	$\mathbb{E}[\mathrm{IC}(F_{t-l}, R_t)]$ vs. lag l	shelf life of the signal
factor turnover	$\mathbb{E}[\mathrm{corr}(F_{t-l}, F_t)]$	trading cost proxy

Purged cross-validation

5 folds of contiguous day blocks, 21-day purge + embargo. Random row splits on overlapping labels are textbook leakage. All tuning on inner chronological slices only.

Paired tests, out of fold

“A beats B” is decided on the out-of-fold daily difference $\mathrm{IC}^A_t - \mathrm{IC}^B_t$: same days, same folds. A method never gets credit for in-sample fit.

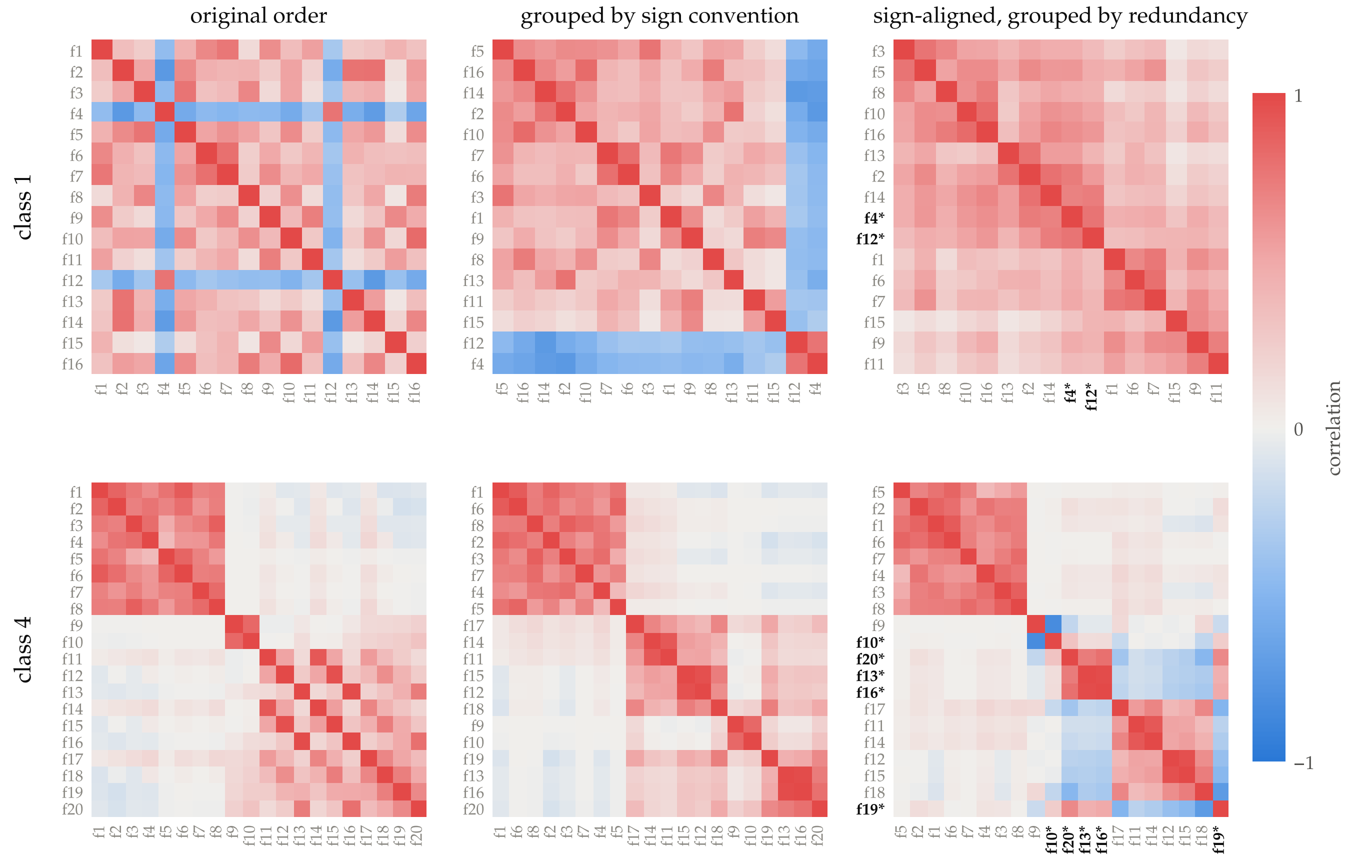
One harness, throughout

Everything that follows runs through this harness unchanged. It will decide half the story, starting with the very first escalation.

FOR THE ML CROWD

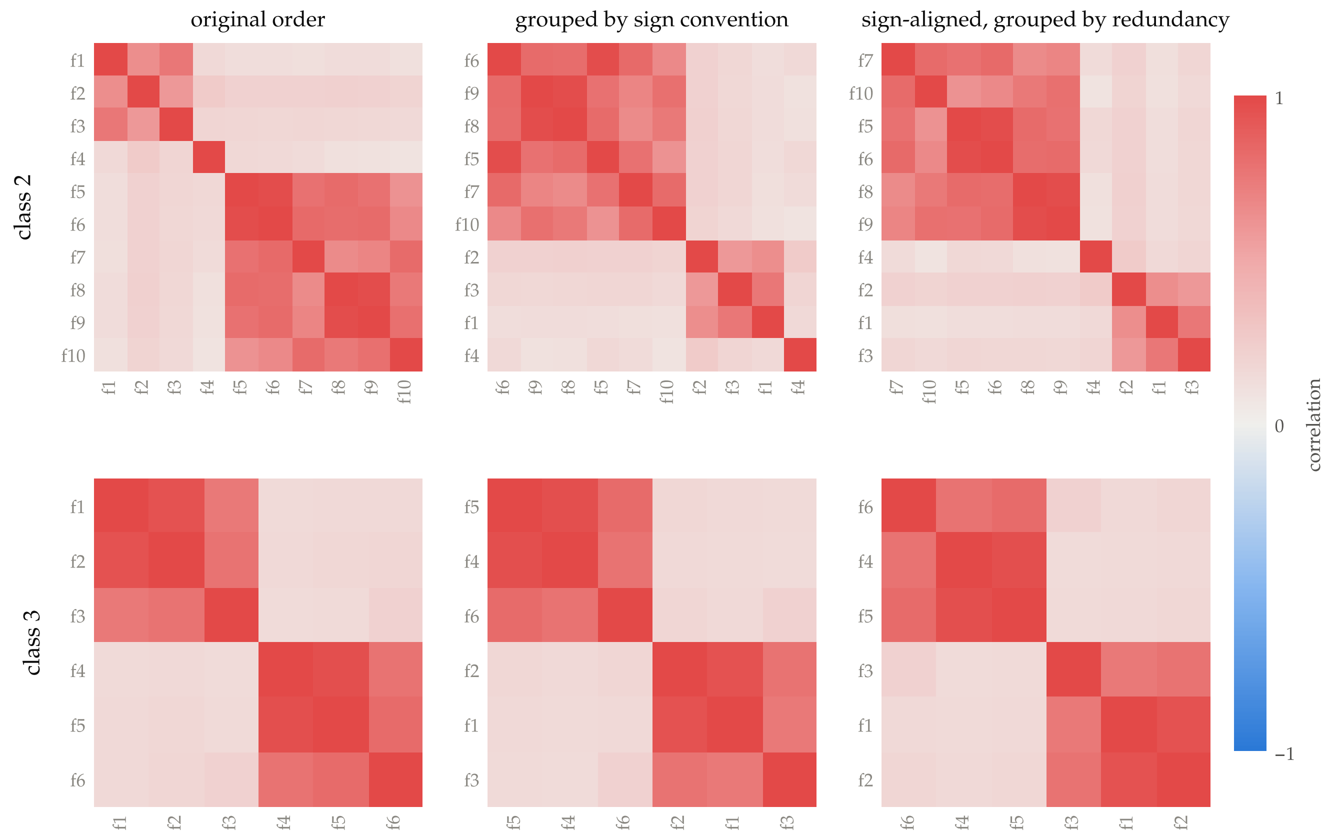
Daily IC = rank correlation of predictions vs realized targets, one number per day; 0.02–0.05 is a strong equity signal. Think “per-day validation metric with autocorrelation-robust error bars.”

Two things hide in a correlation matrix; only one is fixable



Class 1 (top): an orientation artifact. Flip just 2 of 16 variants and the messy grid becomes one coherent block: one theme, inconsistent sign conventions. **Class 4 (bottom): real structure.** Flipping its flagged variants makes things *worse*; the disagreement between its sub-themes is genuine. **And alignment alone is not enough:** class 1's aligned orientation points the *wrong way*, so at least one sign per cluster must come from the target.

Classes 2 and 3 arrive already coherent



The other two classes, same three views: each is **one solid, consistently-signed block**. No orientation artifact to fix, no sub-theme disagreement: essentially one idea measured several ways, in blocks of 0.7–0.95 correlation. **Remember this picture:** with one clean factor there is nothing for estimated weights to exploit, and that is exactly where the supervision tax will land.

THE ROUTE

Eight stages, run in the order the results demanded

1 · TRY

=

Baselines

Benchmark, IC weights, ridge, OLS, the uniqueness GLS. All tie at the ceiling.

2 · ESCALATE

✗

PLS & trees

Built to break the tie. They certify it instead: one predictive direction, no nonlinear edge.

3 · DIAGNOSE

✓

Predictive dimension

Variance dimension ≠ predictive dimension. The map that dictates the design.

4 · THE MOVE

✓

Deduplication

Count each idea once. The only method that never loses, and the study's one win.

5 · ADAPT

✗

Adaptive weights

Meta-analytic shrinkage. Flawless detector; the tilt dies on the blind test.

6 · LEARN

◑

Feature shapes

The one thing trees add, kept behind an evidence gate. Adopted on class 1 alone.

7 · ATTACK

✓

Stress tests

80 perturbed feature sets. The dedup composite holds the shallowest worst case.

8 · FREEZE

0

Blind hold-out

~2,180 unseen days. Signals real, IC edge zero, quality edge real, fragile things fail on schedule.

Stage 1 · Every sensible weighting lands in the same place

TRIED (CLASS 1, PURGED 5-FOLD CV)	MEAN IC
benchmark (rank-and-average)	0.0309
IC weighting	0.0309
ridge regression	0.0308
uniqueness regression (GLS)	0.0306

A band a hair wide, indistinguishable inside fold noise. OLS and lasso land in the same band.

TRANSLATION

The uniqueness regression is GLS with a ridge prior: weights ∝ $(\Sigma+\lambda I)^{-1}\hat{\imath}$. The literal implementation of "weight by uniqueness and information content."

- **What happened:** nothing separates. Within a class the variants' |IC|s are broadly similar, so clever weighting has little cross-sectional signal to exploit, and still pays its estimation noise.
- **The revealing detail:** the uniqueness regression does *better* the more it ignores the estimated correlation matrix. On a near-one-dimensional cluster, the redundancy correction has nothing true to find, so it can only amplify noise.

VERDICT

All tied. Either the class is saturated, or the methods are too weak to show it.

NEXT MOVE

Escalate capacity: supervised directions (PLS) and nonlinearity (trees).

Stage 2 · The escalation built to break the tie certifies it

Tried: PLS, tuned components

How many supervised directions does the cluster hold? The tuner picks **a single component, every time**. One-component PLS *is* IC weighting, by algebra: the tie is confirmed, not broken.

Tried: boosted trees, leak-free

XGBoost's in-sample IC of **0.047** collapses to **0.031** the moment it is scored out of fold, and its remaining edge is a rounding error. Even its own tuner keeps asking for minimal capacity.

THE HARNESS PAYS OFF ON DAY ONE

The 0.047 → 0.031 collapse is exactly why no method gets credit for its own fit. Every "edge" in this study must survive the paired out-of-fold test first.

VERDICT

The class-1 ceiling is real: one predictive direction, no nonlinear structure worth the name.

NEXT MOVE

Run the full suite on all four classes and ask *where* they differ.

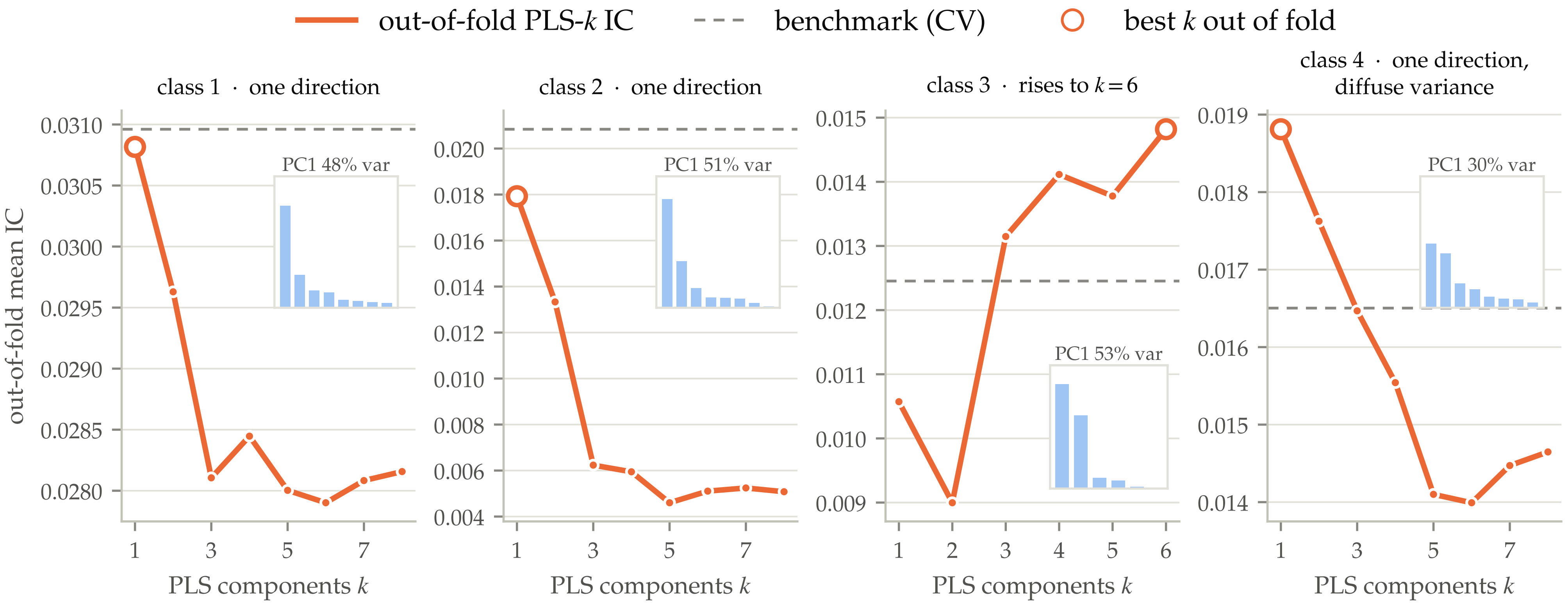
Stage 3 · The map: every class has a ceiling

purged 5-fold CV mean IC · shading scaled within each class,
bold = class best

	class 1	class 2	class 3	class 4
benchmark	0.0309	**0.0208**	0.0124	0.0165
IC weighting	0.0309	0.0173	0.0108	**0.0198**
ridge	0.0308	0.0090	0.0141	0.0160
OLS	0.0281	0.0057	**0.0148**	0.0143
PLS (tuned)	0.0308	0.0133	0.0146	0.0163
uniqueness regression	0.0306	0.0132	0.0114	0.0187
PCA PC1	0.0305	0.0183	0.0125	0.0033
boosted trees	**0.0315**	0.0095	0.0048	0.0121

Purged CV mean IC, every method × every class (darker = higher within a class; bold = class best). **Each class has its own ceiling, and the benchmark sits at or within noise of it everywhere**: it literally wins class 2. Supervision helps only on the diffuse class 4, and only slightly: the flagged protagonist, on cue. **New fact: trees collapse off class 1**, losing a quarter to two-thirds of the benchmark's IC elsewhere (LightGBM reproduces both the parity and the collapse). **Next: the differences between classes aren't about accuracy. Diagnose what they are about.**

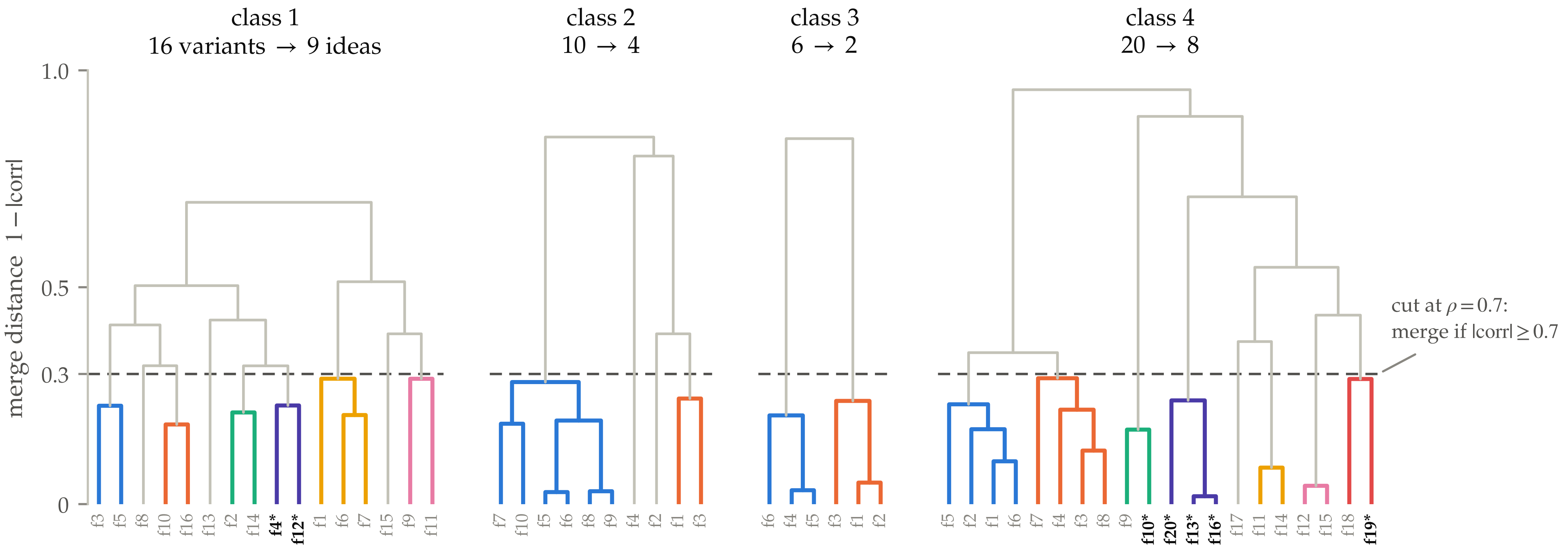
Stage 3 · Variance dimension ≠ predictive dimension



Out-of-fold IC as supervised directions are added (best point circled; benchmark dashed), with each cluster's variance spectrum inset. **Class 4 looks multi-factor but predicts one-dimensionally**: the fix is stopping the most-duplicated definition from drowning the one direction that matters. **Class 3 is the reverse**: its predictive content hides in low-variance directions, a temptation resting on one tuned parameter from the weakest class. File it away; it returns on the hold-out.

Verdict: one route satisfies all four readings at once. Cluster the redundancy away, weight the surviving ideas equally.

Stage 4 · The move: 52 variants → 23 ideas



Cluster the variants on correlation distance; the dashed cut at **ρ = 0.7** merges any variants sharing at least half their variance. Colored subtrees are redundancy blocks (one block = one idea); gray leaves are already unique. **Across the four classes, 52 variants collapse to 23 ideas.** Each block becomes its sign-aligned average, and the *blocks* get the weights. The threshold is **fixed, not tuned**: tuning was offered twice and never beat it. Clustering is the hard form of an equal-weight prior: structure instead of another free parameter.

Stage 4 · The win, and the tax that frames it

The one win: where duplication bias actually binds

Class 4, the one genuinely multi-factor cluster: dedup + equal weights significantly beats the benchmark (**+0.0028** daily IC, t = **2.26**), using **nothing from the target beyond sign bits**. Supervised rivals gain slightly more on paper, with less statistical support: estimation adds gains and variance in equal measure.

The supervision tax: where the cluster is one clean factor

Class 2, one clean factor: *every* weighting that touches the target does significantly *worse* than the benchmark, with deficits up to half the class's IC. The more a method estimates, the more it loses.

FOR THE ML CROWD

The forecast-combination puzzle (DeMiguel's 1/N result) inside a single signal: estimated ensemble weights carry estimation error comparable to the differences they chase, so uniform weights win. The fix that works is *structural*, not another estimated parameter.

VERDICT

Premise confirmed: the benchmark's flaw is duplication, not equal weighting. Dedup is kept.

NEXT MOVE

Equal vs IC-tilted *block* weights is still open. Decide it by estimation, not search.

Stage 5 · Borrowed from medicine: shrinkage that knows when to be equal

META-ANALYSIS	FEATURE COMPOSITION
study *i*	redundancy block *i*
effect estimate θ_i	block's train-window mean IC
within-study s.e.	Newey–West s.e. of that mean
duplicate publications	re-implemented definitions
fixed-effect pooling	≈ IC weighting
forcing equal weights	the benchmark

$$\tilde{\theta}_i = \bar{\theta} + \lambda_i\,(\hat{\theta}_i - \bar{\theta}), \qquad \lambda_i = \frac{\hat{\tau}^2}{\hat{\tau}^2 + se_i^2}$$

FOR THE ML CROWD

Empirical-Bayes / James–Stein shrinkage of per-block means toward the pooled mean. τ^2 is the between-idea variance, estimated in closed form: the shrinkage strength *is* the regularizer, and it is estimated, not tuned.

What happened: a flawless detector

From training data alone, in every fold: the estimated between-idea variance is **zero on classes 1–3**, so equal weights fall out automatically, and **positive on class 4**, where the blocks truly differ. One closed form recovered the whole supervision map, and on class 2 it side-steps the supervision tax by itself.

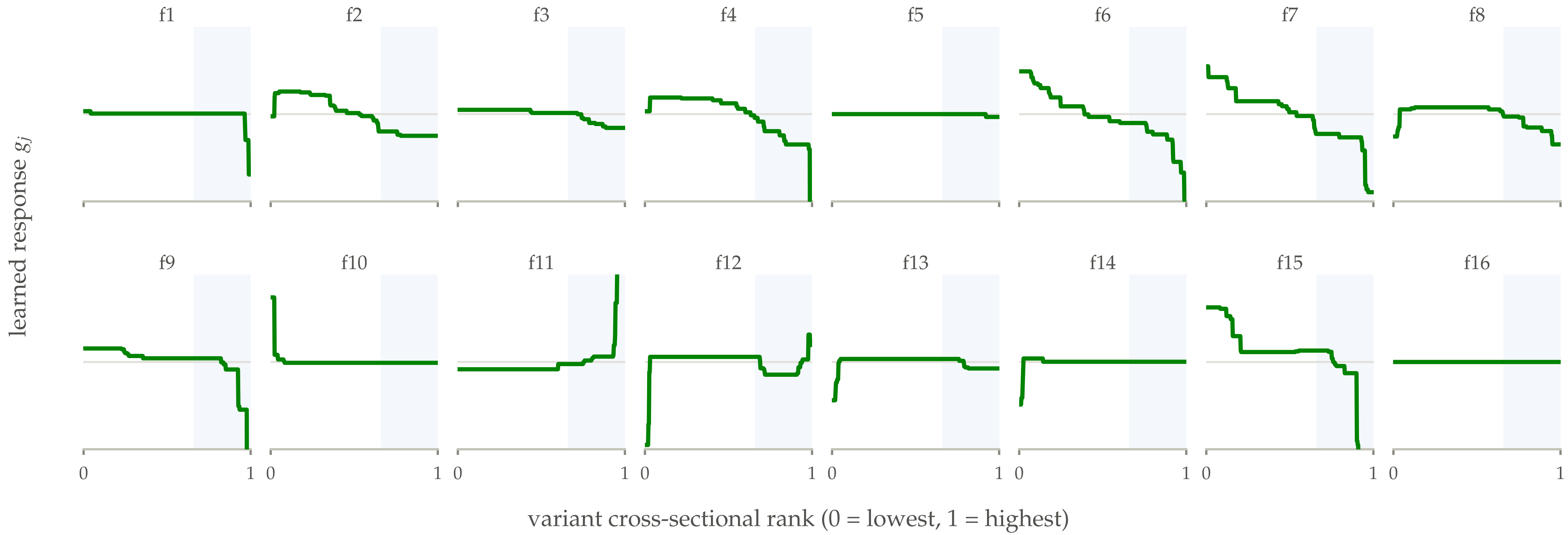
VERDICT

Detector: kept. But detection and monetization are different claims.

NEXT MOVE

The tilt's real test is the blind data. Verdict pending.

Stage 6 · Decoding the trees: the edge is 16 lookup tables



The loose end from stage 2: trees tied on class-1 accuracy, but their signal aged better: nearly double the half-life at visibly lower turnover. The ablation: depth-1 stumps, which *cannot* represent interactions, reproduce the full tree's profile. So the edge is **learned per-feature response curves**: a stump ensemble is exactly an additive model, and the production version is **16 lookup tables**, no tree library required. The curves are flat through mid-ranks, where daily rank jitter lives, and respond only in the extreme tail. **That is the turnover mechanism: the composite stops changing its mind over noise.**

Stage 6 · Extra capacity only enters behind a mechanical gate

Condition A: accuracy non-inferiority

Paired Newey–West *t* of shaped vs unshaped, out of fold under purged CV, must exceed −2.

Condition B: strict tradeability win

5-day IC retention strictly *up* **and** 5-day factor turnover strictly *down*, simultaneously. Shapes must pay where the linear composite actually churns.

Defined from in-sample quantities only; no tuned constants. **The gate is the one data-driven decision point in the final method.**

Adopted: class 1 alone

Shaped dedup composite on class 1: **double the half-life at roughly half the turnover**, at the same accuracy. It out-trades the tree that taught it the shapes.

Refused: classes 2–4

Applied unconditionally, shapes destroy most of class 2’s IC: hundreds of stumps on a short panel is the tree collapse all over again.

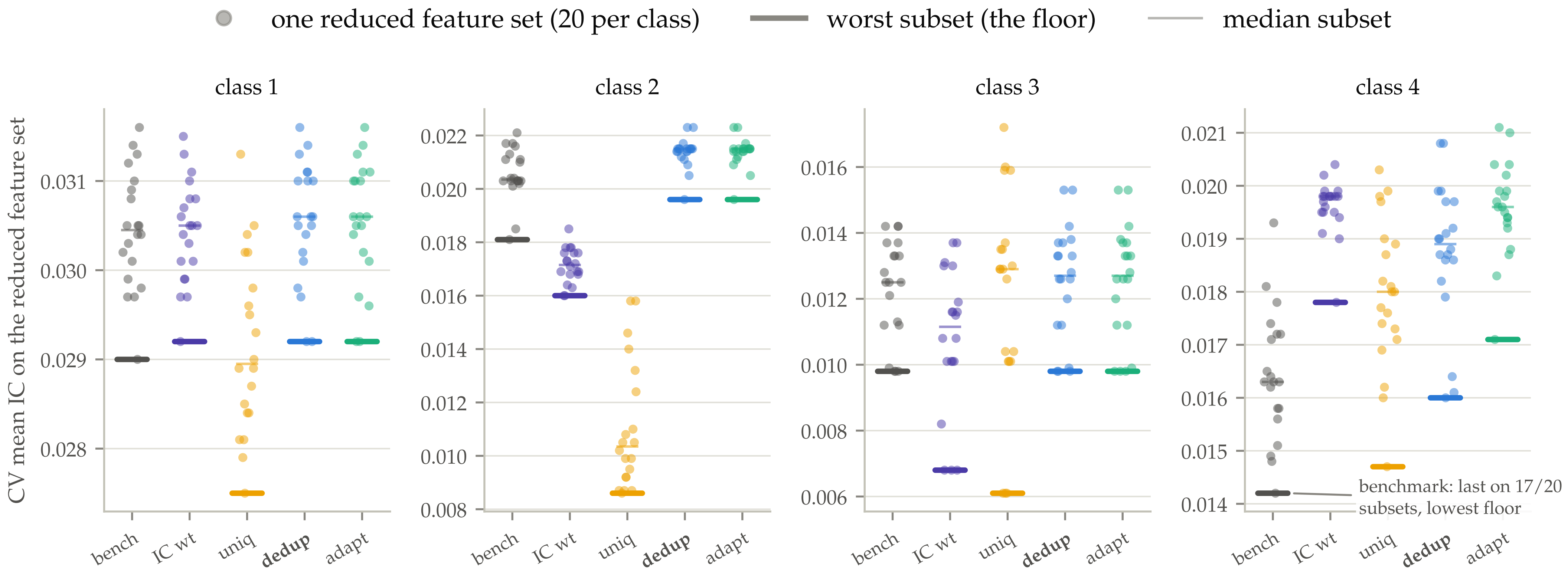
VERDICT

Shapes: conditional, class 1 only.

NEXT MOVE

The composite is fully specified. Attack it before freezing.

Stage 7 · Perturb the feature list: who has the shallowest worst case?



80 randomly reduced feature sets (20 per class, about a quarter of the variants dropped, the full purged CV re-run each time). Heavy tick = each method's worst subset, its floor. **Dedup holds the highest or joint-highest floor on three classes and the shallowest worst case overall.** Everyone else craters somewhere: the benchmark is the casualty on class 4, where dropped variants leave its duplication bias nothing to hide behind, and the tuned methods crater on classes 2 and 3. **Verdict: this robustness is invisible in any single full-set comparison. Next: freeze everything and go blind.**

Stage 8 · Freeze everything, score once

- **The data:** the immediately following ~2,200 trading days for every class, delivered only after the in-sample study was complete.
- **The freeze:** feature signs, IC weights, tuned λ and ρ, the clustering itself, adaptive block weights, the XGBoost model, the stump shapes, the gate decision, even standardization constants. Estimated once in-sample, applied unchanged.
- **Nothing** is fit, tuned, or selected on the hold-out.

Three verdicts ride on this test

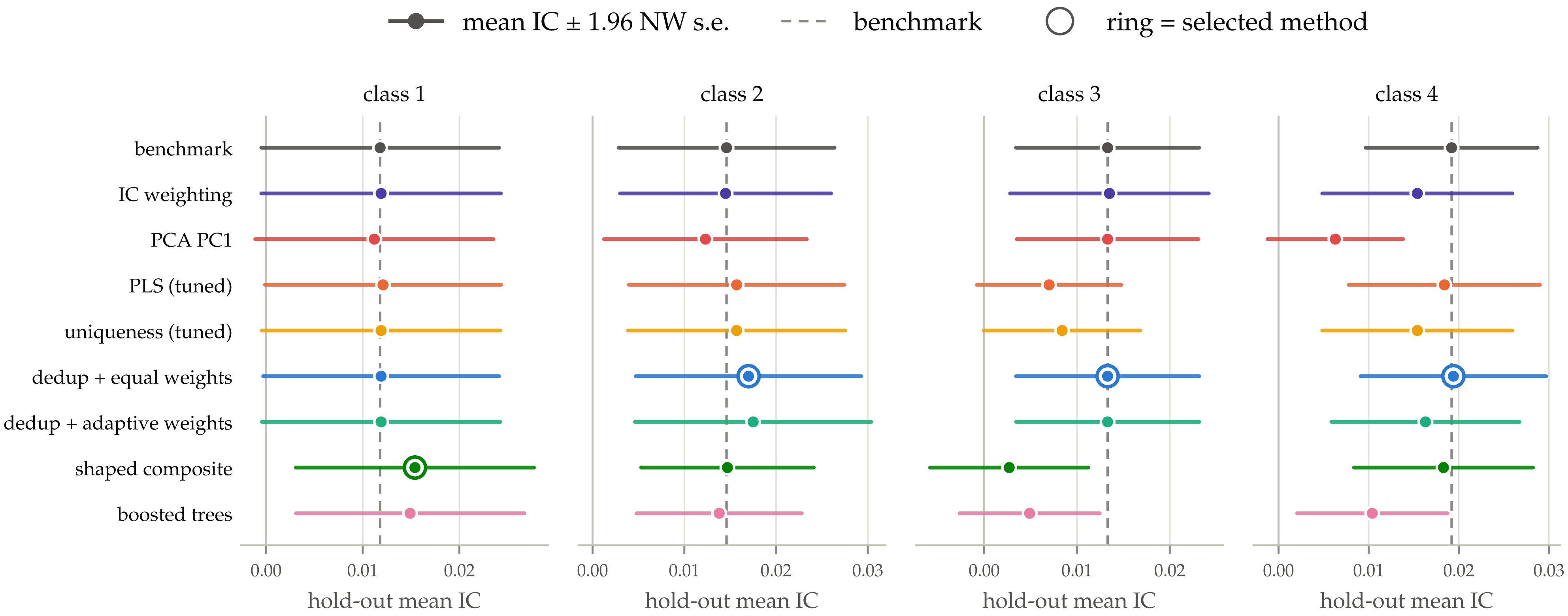
The adaptive tilt (pending), the shapes (conditional), and the accuracy claim itself. Everything the study flagged as fragile is about to be tested on ~2,180 days none of the methods ever saw.

DISCLOSURE

During the final review phase the hold-out was consulted repeatedly; every such use is labeled in the notebook, the specification is frozen as of 2026-07-11, and the next data delivery is the certification event, not another tuning opportunity.

1 BASELINES = › 2 ESCALATE ✗ › 3 DIAGNOSE ✓ › 4 DEDUP ✓ › 5 ADAPT ? › 6 SHAPES ◑ › 7 STRESS ✓ › 8 BLIND

Stage 8 · Real signals, zero IC edge, fragile things fail on schedule



Mean daily IC with 95% confidence whiskers, ~2,200 unseen days per class; dashed line = benchmark, ring = selected method. **The signals are real**: classes 2–4 stay significantly positive under strict freeze (t between 2.4 and 3.9). **The accuracy claim replicates at zero**: the composite is statistically indistinguishable from the benchmark, matching or exceeding it on every class. **And the flagged fragilities failed on schedule**: PCA collapses on class 4; tuned PLS and the shaped arm break on class 3. The next three slides are those casualties, each one a lesson.

Casualty 1 · Detection survived; monetization didn’t

The adaptive tilt, frozen on class 4

On class 4, the one class where its weights differ from equal at all, the tilt **significantly underperforms** plain equal weights (**−0.0031** daily IC, $t = -3.4$). Re-estimating it quarterly does not rescue it: tilt *plus maintenance* still loses to equal weights frozen forever.

But the detector is estimator-robust

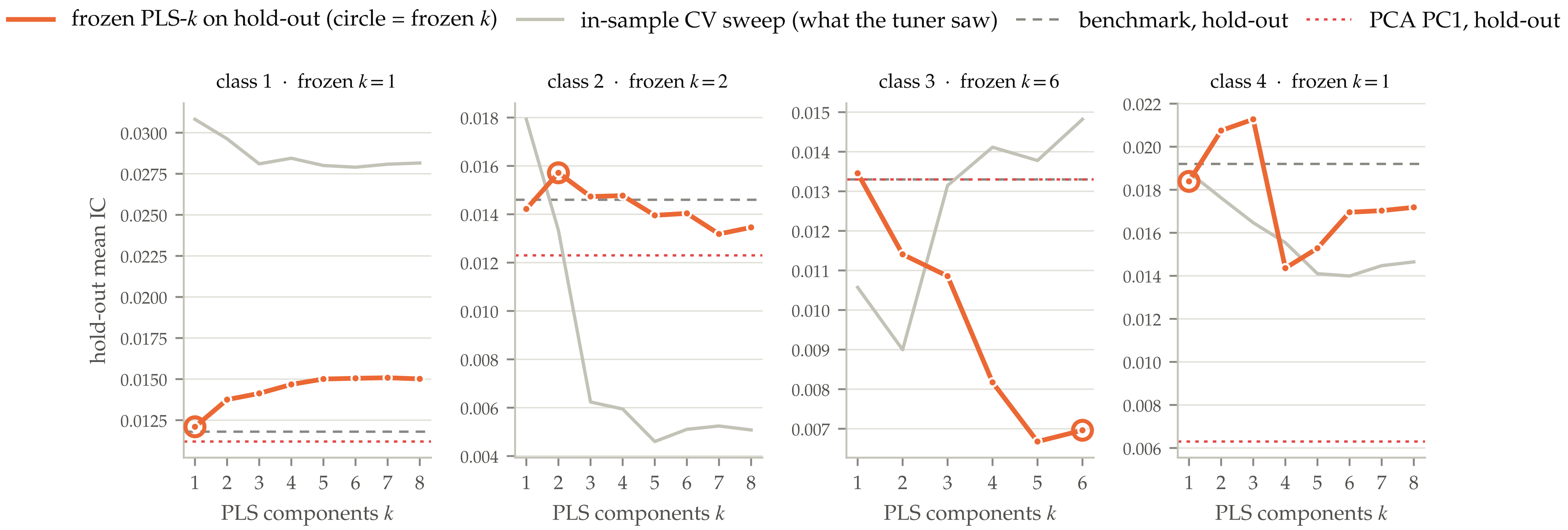
Paule–Mandel and REML reproduce the τ^2 map exactly: zero on classes 1–3, positive on class 4 in every fold, under all three estimators.

WHY, IN ONE NUMBER

With only a handful of blocks per class, the uncertainty band around any block’s true IC is far wider than the differences between blocks. At these block counts the estimator can *detect* heterogeneity but cannot *support* differential weights.

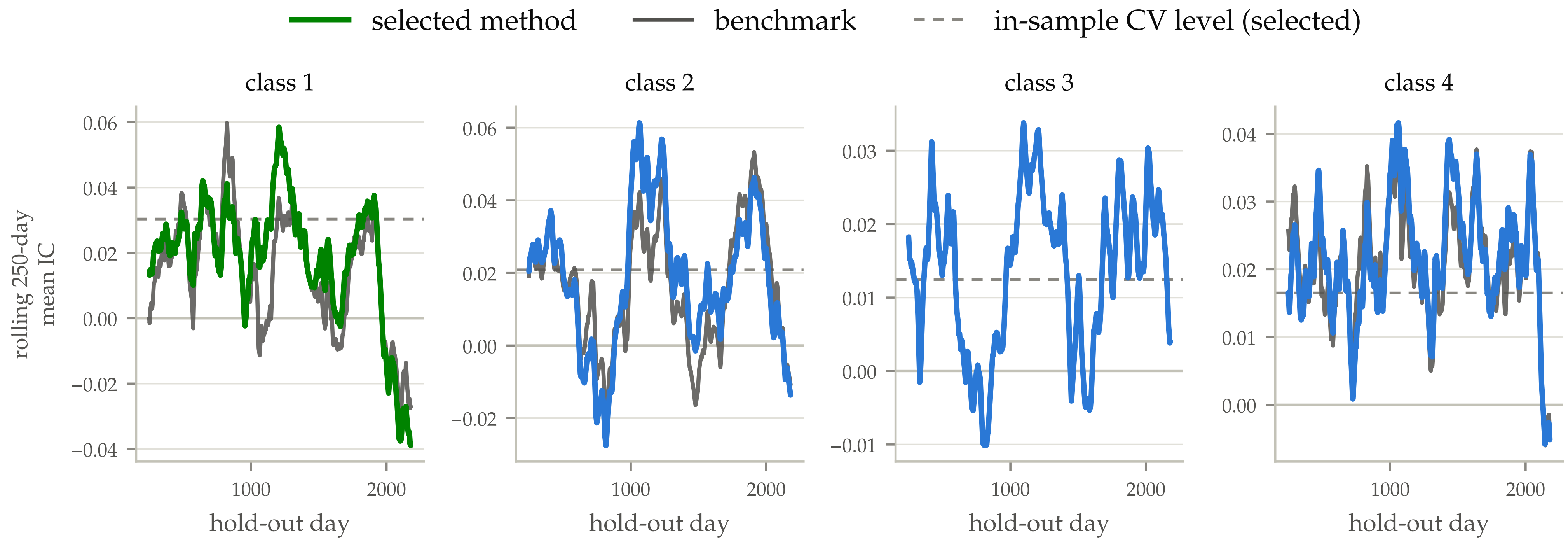
The lesson that shaped the final method: keep τ^2 as a diagnostic flag (“this class is genuinely multi-factor, pay attention”) and act with equal weights anyway.

Casualty 2 · Leak-free tuning that still died: PLS on class 3



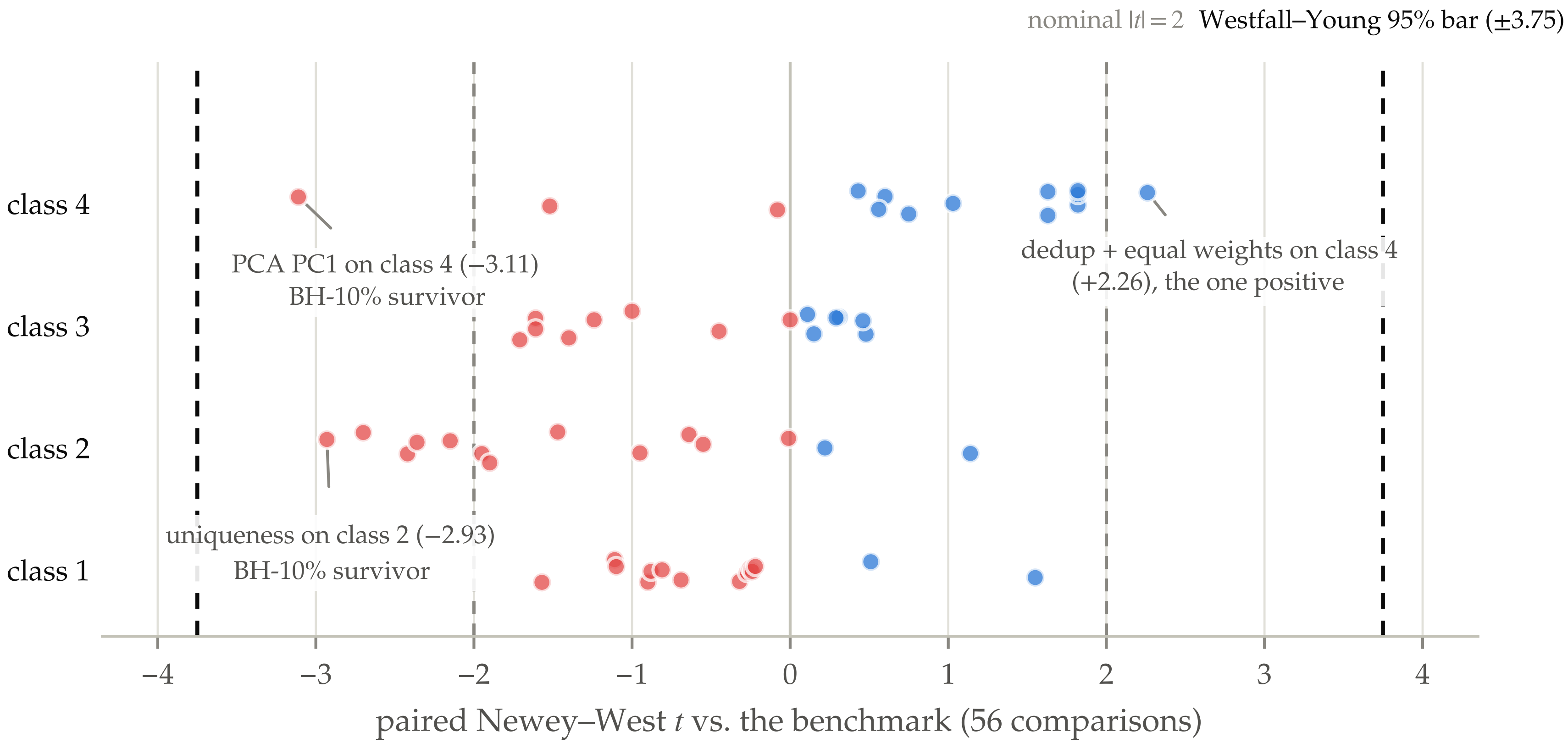
The class-3 temptation from the diagnosis stage, coming home. Orange = frozen PLS on the hold-out as components are added; circle = the choice made leak-free in-sample and then frozen; gray = the in-sample sweep the tuner saw. On class 3 the tuner chose six components, a perfectly defensible call, and **the hold-out preferred one: that single choice cost the class half its IC.** Same lesson as the tuned threshold and the monetized tilt: introduce estimation, and the intended benefit is lost to noise.

Casualty 3 · Class 1 lost two-thirds of its IC; nothing re-weightable was lost



Rolling one-year hold-out IC (green = selected, gray = benchmark, dashed = in-sample level). Class 1 lost **two-thirds of its IC**, concentrated in the final year. Yet the structure barely moved: the correlation matrix is stable and not one of the 16 variants flipped sign. Rolling re-estimation recovers **exactly nothing**, because nothing re-weightable was lost. **Moral for any frozen composite: monitor the level, not just the structure. Alpha decay cannot be recovered by refitting.** Silver lining: the shaped composite is the one class-1 method still clearly significant.

Correct all 56 comparisons: only the negative results survive



Every paired method-vs-benchmark comparison the study ever ran: **56 tests**, one dot per method and class. Nominal significance flags seven. Corrected for the whole family, **zero survive**, including our own headline win. At the loosest standard tried, exactly two results survive, **both negative**: PCA on the multi-factor class, tuned uniqueness on the one-factor class. **So the defensible claim is the floor, not the mean**, and the certified findings are prohibitions: don't chase variance on a multi-factor cluster; don't supervise a one-factor one.

Priced at a weekly horizon, the composite is consistently ahead

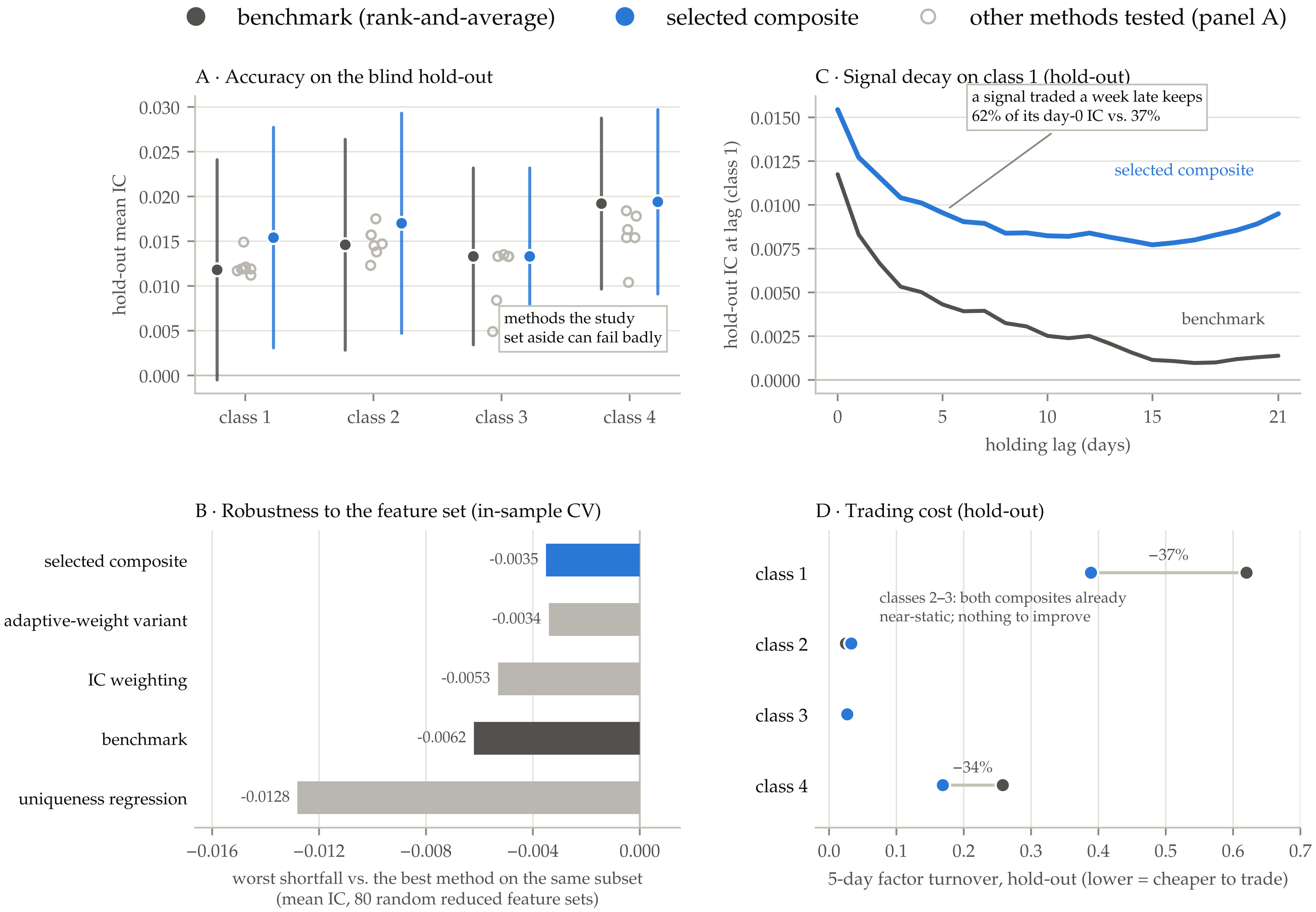
FOUR-CLASS PACKAGE	EDGE/DAY	NW *T*
shaped c1 + dedup-equal c2–4 (selected)	+0.0019	+1.72
shaped c1 + dedup-adaptive c2–4	+0.0013	+1.02
dedup equal weights everywhere	+0.0007	+1.05
dedup adaptive weights everywhere	+0.0001	+0.09
PCA PC1 everywhere (retired ref.)	−0.0039	−3.07
PLS tuned everywhere (retired ref.)	−0.0013	−1.20

Criterion: the average of the fresh and week-old IC, since a weekly book trades a signal that is zero to five days stale; paired daily against the benchmark on the hold-out. The retired references at the bottom show the test has real power to flag a bad package, so the selected edge competes against a real zero. Suggestive, not certified: the package is post hoc, its components pre-specified.

Selected − benchmark, hold-out IC at every holding lag: **at or above zero everywhere**, and growing with the holding period on the two classes where the method changed anything, at 35–40% lower turnover.

RECAP

The study in one figure



THE SELECTED COMPOSITE

Four steps, no tuned parameters, frozen

- **1 · Group the duplicates.** Cluster on 1−|corr| (average linkage); merge above |corr| ≥ 0.7. Same idea, several implementations, one vote.
- **2 · Average within blocks.** Flip negative-IC variants, then simply average each block's members.
- **3 · Weight ideas equally, freeze.** The tilt detected real heterogeneity; acting on it lost. τ^2 stays as a diagnostic, never as weights.
- **4 · Shapes only past the gate.** Per-feature response curves where the two-stage bar clears: empirically class 1 alone.

CLASS	COMPOSITE	IC	NW *T*	5D RET.	5D TURN.
1	shaped → dedup → equal blocks	0.0154	2.45	0.62	0.39
2	raw → dedup → equal blocks	0.0170	2.71	0.94	0.03
3	raw → dedup → equal blocks	0.0133	2.64	0.97	0.03
4	raw → dedup → equal blocks	0.0194	3.69	0.89	0.17

In one sentence: count each idea once, weight ideas equally, learn feature shapes only where the evidence is strong, and freeze everything.

Deliberately left out, each with a documented failure

IC weighting (taxes one-factor clusters) · GLS and uniqueness solves (inverted out of sample) · the adaptive tilt (lost on the blind test) · scheduled re-estimation (repairs nothing) · boosted trees off class 1 (collapse) · tuned thresholds and method selection (whipsaw) · PCA (the wrong objective). **Adopting the "obvious" improvements without this discipline lands strictly below plain rank-and-average.**

A playbook for the next feature class

- **1 · PC1 share + the dendrogram.** Above ~0.45 with clean blocks: equal weighting will saturate the class. Diffuse (~0.3): the partition matters; expect the plain benchmark to be the casualty.
- **2 · The out-of-fold PLS-*k* sweep.** Peak at k=1: any sensible weighting works. Rising past $k \geq 3$: real multi-directional signal. Treat as a research flag, not a production decision (both frozen fits inverted out of sample).
- **3 · τ^2 under all three estimators.** Zero: equal weights are optimal-in-family; walk away. Positive under all three: the blocks truly differ; with $k \geq 20$ blocks or known block identities, the tilt deserves a fresh trial.
- **4 · Run the gate for shapes.** Its measured record: 19 of 20 correct calls and one prevented out-of-sample blow-up. The conservative failure mode you want in production.
- **5 · After deployment: two separate alarms.** Track the trailing-year mean IC against its in-sample range, *and* structural drift as a second, independent alarm. Class 1 proved a drift alarm alone misses the failure that actually happens.

> **WHY THIS IS THE DELIVERABLE**
> Four classes can't certify a universal rule. What they can certify is a set of cheap mechanical diagnostics with a measured track record, plus a map from readings to actions.

Where this goes next

Compose the composites

The four selected composites are themselves four correlated, noisy signals: exactly the structure this study composes. Run the identical method one level up, where block identities are known and economically meaningful.

Un-blind the classes

Every diagnostic ran blind. Knowing the theme converts the map into interpretation: does class 1's uniform alpha decay match that theme's crowding cycle? The shapes are lookup tables a PM can read directly.

More classes → validate the rule itself

The diagnose-then-act rule has validated components but four end-to-end trials. Twenty classes would cross-validate the rule, and would re-try the two ideas we could detect but not reward (τ^2 tilt, feature-level GLS) exactly where the diagnostics say they should work.

Real cost models

Retention, turnover, half-life are cost proxies. With your transaction-cost model the weekly criterion becomes net-of-cost P&L, and "half the turnover at equal IC" on class 1 converts into a directly bankable number.

THANK YOU

Count each idea once, weight ideas equally, learn shapes only on strong evidence, and freeze everything.

Not more IC. Nothing in the study delivers that, and we can prove it. What the composite beats is the benchmark's **fragility**: its duplication bias, its churn, and its false promise that sophistication is free.

Full study: 200-cell reproducible notebook + 28-page report.